

Prioritised Revision Guide

Cambridge Part II Statistical Mechanics

Based on analysis of all 15 past papers (2011–2025)
and the course handout (Terentjev)

How to use this guide

Topics are ranked by exam frequency, weighted towards the most recent papers (2019–2025). Each entry specifies what to derive, what to sketch, and which years the topic appeared. The **green percentages** show the fraction of papers in which the topic appeared. Prepare Tier 1 first, then Tier 2, etc.

TIER 1 — Near-Certain to Appear (80%+ frequency)

1. Heat Capacity Derivations

93%

The single most examined skill. You must compute C from a partition function or density of states for any system.

Two-level system (Schottky anomaly)

$Z = 1 + e^{-\beta\epsilon}$. Internal energy $U = \epsilon e^{-\beta\epsilon}/Z$. Heat capacity peaks at $k_B T \approx 0.42 \epsilon$, vanishes at both $T \rightarrow 0$ and $T \rightarrow \infty$. **Sketch** $U(T)$ and $C(T)$.

Tested: 2013, 2014, 2020, 2021, 2024, 2025

Quantum harmonic oscillator

$Z = e^{-\beta\hbar\omega/2}/(1 - e^{-\beta\hbar\omega})$. High- T : $C \rightarrow k_B$. Low- T : $C \propto (\hbar\omega/k_B T)^2 e^{-\hbar\omega/k_B T}$.

Tested: 2014

Ideal Fermi gas at low T

$$C = \gamma T, \quad \gamma = \frac{\pi^2}{3} k_B^2 g(\epsilon_F) \quad (1)$$

Derive by Sommerfeld expansion or by the “smearing” argument: only electrons within $\sim k_B T$ of ϵ_F contribute.

Tested: 2014, 2015, 2016, 2018, 2019, 2023, 2025

Debye model (phonons)

$C \propto T^3$ at low T ; $C \rightarrow 3Nk_B$ at high T . Know Debye temperature Θ_D , density of states $g(\omega) \propto \omega^2$ up to ω_D .

Tested: 2018, 2020

Generalised dispersion $\varepsilon = A|k|^\alpha$ in d dimensions**Key formula — the examiners' favourite device**For a system with dispersion $\varepsilon \propto |k|^\alpha$ in d dimensions:

$$g(\varepsilon) \propto \varepsilon^{d/\alpha-1}, \quad U \propto T^{d/\alpha+1}, \quad C \propto T^{d/\alpha}$$

For bosons with $\mu = 0$ (non-conserved quasiparticles). For conserved fermions at low T : $C \propto T$.*Tested: 2012, 2017, 2018, 2019, 2020, 2022, 2023, 2025***General technique**Given *any* new system: (1) write down $g(\varepsilon)$ or Z , (2) compute $U = -\partial \ln Z / \partial \beta$ or $U = \int \varepsilon g(\varepsilon) \bar{n}(\varepsilon) d\varepsilon$, (3) differentiate to get C .**2. Density of States for Arbitrary Systems****60%**

Not usually standalone, but underpins almost every long question.

Free particles in d dimensions, $\varepsilon = A|k|^\alpha$: Count states in a d -dimensional k -shell: number of states $= S_d k^{d-1} dk / (2\pi)^d$. Change variables to ε :

$$g(\varepsilon) \propto \varepsilon^{d/\alpha-1}$$

Standard cases to memorise:

System	Dispersion	Dimension	$g(\varepsilon)$
Free particle	$\varepsilon \propto k^2$	3D	$\propto \varepsilon^{1/2}$
Free particle	$\varepsilon \propto k^2$	2D	const
Linear (photon/phonon)	$\varepsilon \propto k$	3D	$\propto \varepsilon^2$
Linear	$\varepsilon \propto k$	2D	$\propto \varepsilon$
Linear	$\varepsilon \propto k$	1D	const
Harmonic trap	—	d D	$\propto \varepsilon^{d-1}$

Semi-classical (Thomas–Fermi) approach for particles in a potential $U(r)$: $g(\varepsilon) \propto \int_0^{r_\varepsilon} r^{d-1} [\varepsilon - U(r)]^{d/2-1} dr$.*Tested: 2016, 2025***3. Bose–Einstein Condensation 53% explicit; ~80% including related content****Standard 3D free BEC****Sketch $\mu(T)$:** starts negative at high T , rises to 0 at T_c . Below T_c : $\mu = 0$ and

$$\frac{N_0}{N} = 1 - \left(\frac{T}{T_c} \right)^{3/2}, \quad T_c \propto \frac{\hbar^2}{mk_B} n^{2/3}$$

BEC condition for general $\varepsilon \propto |k|^\alpha$ in d dimensions**BEC criterion**BEC occurs $\iff \int_0^\infty \frac{g(\varepsilon)}{e^{\beta\varepsilon} - 1} d\varepsilon$ converges (i.e. N_{excited} is bounded).Convergence requires $d/\alpha > 1$. Examples: 3D free gas ($d/\alpha = 3/2$): $\checkmark$. 2D free ($d/\alpha = 1$): borderline, no BEC. 1D: never.*Tested: 2019, 2022, 2025*

BEC in harmonic trap

$g(\varepsilon) \propto \varepsilon^{d-1}$. BEC occurs for $d > 1$ (so $d \geq 2$ in a trap, vs $d \geq 3$ free). $T_c \propto \hbar\omega N^{1/d}/k_B$.

Tested: 2016, 2025

2D free Bose gas — no BEC

$\mu = k_B T \ln(1 - e^{-T_0/T})$ where $T_0 = 2\pi\hbar^2 n/(mk_B)$. Chemical potential never reaches 0.

Tested: 2022

Photon/phonon gas

Bosonic excitations with $\mu = 0$: $U \propto T^4$, $pV = U/3$.

Tested: 2013, 2020, 2024

4. Virial Expansion & Non-Ideal Gases**53%**

Equation of state: $\frac{pV}{Nk_B T} = 1 + B_2(T)\frac{N}{V} + B_3(T)\left(\frac{N}{V}\right)^2 + \dots$

Second virial coefficient:

$$B_2(T) = -\frac{1}{2} \int [e^{-\beta\varphi(r)} - 1] d^3r \quad (2)$$

Evaluate for hard spheres ($B_2 = 2\pi a^3/3$), square well, and any piecewise potential.

Boyle temperature: T^* where $B_2(T^*) = 0$. At T^* , attractive and repulsive contributions cancel.

Tested: 2019, 2022

Quantum correction: $B_2^{\text{quantum}} = \mp \lambda^3/2^{5/2}$ (+ fermions, − bosons). Physical interpretation: Pauli exclusion gives effective repulsion; bosonic bunching gives effective attraction.

Tested: 2018

5. Canonical Ensemble & Partition Functions**60%**

Derivation: Maximise $S = -k_B \sum_i p_i \ln p_i$ subject to $\langle E \rangle = U$ and $\sum p_i = 1$. Result: $p_i = e^{-\beta E_i}/Z$.

Tested: 2024

Extracting thermodynamics: $F = -k_B T \ln Z$, $S = -\partial F/\partial T$, $p = -\partial F/\partial V$, $U = -\partial \ln Z/\partial \beta$, $C_V = \partial U/\partial T$.

N-particle systems: Distinguishable: $Z_N = Z_1^N$. Indistinguishable classical: $Z_N = Z_1^N/N!$.

Ideal gas: $Z_1 = V/\lambda^3$ where $\lambda = \sqrt{2\pi\hbar^2/(mk_B T)}$. Derive $pV = Nk_B T$ and $U = \frac{3}{2}Nk_B T$.

Particles in traps: $Z_1 = (1/\lambda^3) \int e^{-\beta V(r)} d^3r$. For $V = ar$: $Z_1 \propto T^{9/2}$. For $V = \frac{1}{2}m\omega^2 r^2$: $Z_1 \propto T^3$.

Tested: 2017, 2024

6. Landau Theory & Phase Transitions**53%**

Landau expansion:

$$F(M) = F_0 + a(T - T_c)M^2 + bM^4 - MB$$

Even powers only at $B = 0$ (symmetry $M \rightarrow -M$). $a > 0$, $b > 0$.

Second-order transition ($b > 0$): Below T_c : $M_0 = \pm\sqrt{a(T_c - T)/(2b)}$. Minimum free energy: $F_{\min} = F_0 - a^2(T_c - T)^2/(4b)$.

$$\text{Heat capacity jump: } \Delta C = \frac{a^2 T_c}{2b}$$

Susceptibility: $\chi = 1/[2a(T - T_c)]$ for $T > T_c$.

First-order transition: Occurs when cubic term (φ^3) or $b < 0$ with positive M^6 term is present. **Sketch** $F(M)$ showing metastable minima.

Tested: 2011, 2015

Tricritical point: $b \rightarrow 0$, include $d' M^6$ ($d' > 0$). Below T_c : $F_{\min} = F_0 - 2b^{3/2}(T_c - T)^{3/2}/\sqrt{27d'}$. Critical exponent $\alpha = 1/2$.

Tested: 2018

Critical exponents (mean-field): $\beta = 1/2$ (order parameter), $\gamma = 1$ (susceptibility), $\alpha = 0$ (heat capacity jump).

7. Mean-Field Theory & Ising Model

Rising: 4 of last 5 years

Trend alert

This topic has appeared in 2021, 2022, 2023, and 2025. Treat as near-certain for future papers.

Ising Hamiltonian: $H = -J \sum_{\langle ij \rangle} \sigma_i \sigma_j - m_0 B \sum_i \sigma_i$, $\sigma_i = \pm 1$.

Mean-field approximation: Replace $\sigma_j \rightarrow \langle \sigma \rangle$. Effective single-spin: $H_{\text{eff}} = -(m_0 B + zJ\langle \sigma \rangle)\sigma$, where z = coordination number.

Self-consistency:

$$\langle \sigma \rangle = \tanh\left(\frac{m_0 B + zJ\langle \sigma \rangle}{k_B T}\right), \quad k_B T_c = zJ \quad (3)$$

Connection to Landau: Expand $F = -Nk_B T \ln Z_1 + \frac{1}{2}zJN\langle \sigma \rangle^2$ in powers of $m = m_0\langle \sigma \rangle$ to recover $F = \text{const} + \alpha(T - T_c)m^2 + \beta m^4 - mB$.

Tested: 2022

Hysteresis: Sketch the cubic $B = 2\alpha(T - T_c)m + 4\beta m^3$. For $T < T_c$: $\Delta B \propto |T - T_c|^{3/2}$.

Tested: 2022

1D Ising model (exact): $Z = [2 \cosh(\beta J)]^{N-1}$. Correlation: $\langle \sigma_a \sigma_{a+r} \rangle = [\tanh(\beta J)]^r$.

Tested: 2023

Lattice gas mapping: $n_\alpha = (1 + \sigma_\alpha)/2$ maps occupation numbers onto Ising spins.

Tested: 2021

TIER 2 — High Probability (40–55% frequency)

8. Fermi–Dirac Distribution & Fermi Gas

53%

Derivation: Grand canonical for a single level: $\Xi = 1 + e^{\beta(\mu - \varepsilon)}$, giving $\langle n \rangle = 1/(e^{\beta(\varepsilon - \mu)} + 1)$.

Fermi energy in d dimensions with $\varepsilon \propto k^2$: $\varepsilon_F = (\hbar^2/2m)(2\pi n/\Omega_d)^{2/d}$. 2D: $\varepsilon_F = 2\pi\hbar^2 n/m$.

Low- T chemical potential: $\mu \approx \varepsilon_F \left[1 - \frac{\pi^2}{6}(k_B T)^2 g'(\varepsilon_F)/g(\varepsilon_F)\right]$.

Grand potential: $\Phi = k_B T \int g(\varepsilon) \ln[1 - n_F(\varepsilon)] d\varepsilon$. $pV = -\Phi$. 3D free gas: $pV = \frac{2}{3}U$.

Pauli susceptibility: $\chi = \mu_B^2 g(\varepsilon_F)$.

Tested: 2019

9. Grand Canonical Ensemble

47%

$\Xi = \sum_{N=0}^{\infty} Z_N e^{\beta\mu N} = \prod_k \Xi_k$. Grand potential: $\Phi = -k_B T \ln \Xi = U - TS - \mu N$.

Classical limit: $\Xi_k \approx 1 + e^{-\beta(\varepsilon_k - \mu)}$ when $e^{\beta\mu} \ll 1$. Gives $N/V = e^{\beta\mu}/\lambda^3$, hence $\mu = k_B T \ln(n\lambda^3)$.

Application — Langmuir adsorption: $\Xi_{\text{site}} = 1 + z_s e^{\beta(\mu + \varepsilon)}$. Fractional coverage θ as function of p .

Tested: 2015

Application — defects with Coulomb repulsion: 0, 1, or 2 electrons on a level with interaction U .

Tested: 2013

10. Van der Waals Equation & Critical Phenomena

27% standalone

Derivation: Hard-core $\rightarrow V \rightarrow V - Nb$. Mean-field attraction $\rightarrow$ adds $-a(N/V)^2$. Result: $p = Nk_B T/(V - Nb) - a(N/V)^2$.

Critical point: $V_c = 3Nb$, $k_B T_c = 8a/(27b)$, $p_c = a/(27b^2)$.

Expansion near critical point: With $\varphi = 1 - V/V_c$ and $t = T/T_c - 1$:

$$\frac{p - p(\mathcal{V}_c, t)}{p_c} = 6\varphi t + \frac{3}{2}\varphi^3$$

At coexistence: $\varphi \propto (-t)^{1/2}$, so $\beta = 1/2$ — same as Ising mean-field.

Tested: 2024

Sketch isotherms: $p(V)$ for $T > T_c$, $T = T_c$, $T < T_c$. Label gas, liquid, coexistence, spinodal, critical point.

11. Brownian Motion & Langevin Equation

40%

Langevin equation: $m dv/dt = -\gamma v + \xi(t)$, with $\langle \xi(t) \rangle = 0$, $\langle \xi(t)\xi(t') \rangle = \Gamma \delta(t - t')$.

Fluctuation–dissipation: $\Gamma = 2\gamma k_B T$. Derive by demanding $\langle \frac{1}{2}mv^2 \rangle \rightarrow \frac{1}{2}k_B T$ at long times.

Tested: 2025

Einstein diffusion: $\langle x^2 \rangle = 2Dt$ (1D), $\langle r^2 \rangle = 6Dt$ (3D), $D = k_B T/\gamma$. Velocity relaxation: $\tau = m/\gamma$.

Numerical estimates: Given particle diameter d : $\gamma \approx 3\pi\eta d$ (Stokes). Estimate D , τ , v_{rms} .

Tested: 2011, 2022, 2024

In a potential: Steady-state $\langle v^2 \rangle = \Gamma/(2m\gamma) = k_B T/m$ regardless of confining potential.

Tested: 2025

12. Two-Level Systems & Paramagnetism**47% combined****Spin- $\frac{1}{2}$ in field B :** $Z_1 = 2 \cosh(\beta \mu_B B)$. $M = N \mu_B \tanh(\beta \mu_B B)$.**General spin- J :** $Z_1 = \sinh[(2J+1)x/2] / \sinh(x/2)$, $x = g \mu_B B / (k_B T)$. High- T : Curie law $\chi = C/T$.**Entropy:** S depends only on $B/T \Rightarrow$ adiabatic demagnetisation cooling.*Tested: 2012, 2016***TIER 3 — Moderate Probability (20–35% frequency)****13. Chemical Equilibrium & Reactions****27%**Chemical potential: $\mu = k_B T \ln(N/Z_1)$. Equilibrium: $\sum \nu_i \mu_i = 0$. Law of mass action: $\prod N_i^{\nu_i} = \prod (Z_1^{(i)})^{\nu_i}$. Dissociation increases with T and decreases with density.*Tested: 2016, 2021, 2025***14. Thermodynamic Potentials & Maxwell Relations****33%** $F = U - TS$, $G = U - TS + pV$, $\Phi = U - TS - \mu N$. Maxwell relations from cross-derivatives. Gibbs–Duhem: $d\mu = -s dT + v dp$.*Tested: 2015, 2019, 2023***15. Thermal Fluctuations****27%** $\langle (\Delta x)^2 \rangle = k_B T / (\partial^2 A / \partial x^2)$. Energy: $\langle (\Delta U)^2 \rangle = k_B T^2 C_V$. Capacitor charge: $\langle (\Delta Q)^2 \rangle = k_B T C$, so $V_{\text{rms}} = \sqrt{k_B T / C}$.*Tested: 2013, 2015, 2019, 2023***16. Vacancies in Crystals****13%, but appeared 2017 & 2023**At zero pressure: $n = N e^{-\varepsilon/k_B T}$. At pressure p : $n = N e^{-(\varepsilon+pv)/k_B T}$. Charged crystal (Schottky defects): charge neutrality constrains vacancy numbers.*Tested: 2017, 2023***17. Equipartition Theorem****20%**Each quadratic degree of freedom contributes $\frac{1}{2} k_B T$. Proof: $\langle x_i \partial H / \partial x_j \rangle = \delta_{ij} k_B T$. Fails when $k_B T \ll \hbar \omega$.*Tested: 2013, 2019, 2025***18. Photon & Phonon Gases****40%**Photons: $\varepsilon = \hbar c k$, $\mu = 0$, $U/V \propto T^4$, $pV = U/3$. Debye phonons: $C \propto T^3$ (low T), $C \rightarrow 3N k_B$ (high T). $pV = U/3$ proof by rescaling.*Tested: 2013, 2020, 2022, 2024***TIER 4 — Occasional Topics (<20% frequency)**

19. Clausius–Clapeyron & Phase Boundaries

$dp/dT = L/(T\Delta V)$. Superconductor: $L = -\mu_0 V T H_c dH_c/dT$.

Tested: 2012, 2020

20. Langmuir Adsorption

Grand partition function per site. $\theta = p/(p + p_0(T))$.

Tested: 2015

21. Random Walks & Diffusion

1D walk probability. Continuum limit $\rightarrow$ Fokker–Planck. Characteristic time scales.

Tested: 2021

22. Lattice Gas $\leftrightarrow$ Ising Mapping

$n_\alpha = (1 + \sigma_\alpha)/2$. Liquid–gas transition $\equiv$ ferromagnetic transition.

Tested: 2021

Cross-Cutting Skills (tested every year)

Five skills that earn marks across all questions

- Order-of-magnitude estimates.** At least one per paper. Practise: de Broglie wavelengths, diffusion constants, relaxation times, fluctuation amplitudes, and $k_B T$ at room temperature ($\approx 1/40$ eV $\approx 4 \times 10^{-21}$ J).
- Sketching.** Be prepared to sketch: $\mu(T)$ for BEC, $F(M)$ for Landau theory at various T and B , van der Waals isotherms, $C(T)$ for two-level systems, $\langle n(\varepsilon) \rangle$ for Fermi–Dirac at various T .
- Changing variables in integrals.** Rescale to dimensionless variables ($x = \beta\varepsilon$), convert k -integrals to ε -integrals via the density of states, use standard results:

$$\int_0^\infty \frac{x^n dx}{e^x - 1} = \begin{cases} \pi^2/6 & n = 1 \\ 2.404 & n = 2 \\ \pi^4/15 & n = 3 \end{cases}$$

- Recognising disguised models.** Examiners dress familiar physics in unfamiliar clothing (triplons as bosons, polymer chains as partition functions, lattice gas as Ising model). Strip away the narrative and identify the underlying statistical mechanics.
- Thermodynamic identities on demand.** Be fluent in: $p - T(\partial p/\partial T)_V = -(\partial U/\partial V)_T$; $dG = -S dT + V dp + \mu dN$; $\langle (\Delta x)^2 \rangle = k_B T/(\partial^2 A/\partial x^2)$.